

Math 166

Names: _____

New series from old

Section Number: _____

1. Use the Maclaurin series for $\frac{1}{1-x}$ to find the Maclaurin series for $\frac{1}{(1-x)^3}$. (Hint: differentiate twice!) What is the radius of convergence?

2. Find the 3rd order Taylor polynomial $T_3(x)$ centered at $x = 1$ for $f(x) = \sqrt{x}$ using the Taylor series formula: $\sum_{n=0}^{\infty} \frac{f^{(n)}(1)}{n!} (x-1)^n$.

3. Estimate $\sqrt{1.1}$ using $T_3(x)$ from the previous problem.

4. Find an error bound for the approximation in the previous problem by using the formula $\text{Error}(T_n(x)) = |f(x) - T_n(x)| \leq K \frac{|x-c|^{n+1}}{(n+1)!}$, where K is such that $|f^{(n+1)}(u)| \leq K$ for all u between the center c and x .

5. What is the true error $|f(1.1) - T_3(1.1)|$ for the estimate in question 3?